

When ECC Energy Ordering Depends on Activity: A Matched Post-Route Study

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Abstract—Dynamic-power reports are conditional on the activity used to stimulate a realized circuit. We ask whether an error-correcting-code (ECC) energy ordering survives replacement of vectorless post-route estimation by operation-specific activity while the routed implementation and extracted parasitics remain fixed. Conventional Hamming-style and Hsiao single-error-correcting, double-error-detecting (SECDED) logic are routed at five matched deterministic seeds and a 10-ns target; all ten runs meet timing. Final-gate-netlist activity covers idle, clean write, clean read, single-bit correction, and double-bit detection at the routed-ECC-logic boundary. Vectorless analysis favors Hsiao in 5/5 seeds, but operation-specific energy favors it in only 4/23 matched operation/seed comparisons. Mean Hsiao-minus-conventional deltas are -0.0308 pJ/clean read, $+0.1207$ pJ/write, $+0.3095$ pJ/correction, and $+0.1935$ pJ/detection. Internal and net-switching components oppose one another in several read modes. Compact exact-equivalence and feasibility controls show why hardware structure, implementation condition, and target admission were frozen before substituting activity. The result is not an architecture ranking: under the evaluated conditions, the ordering depends on the stated activity and measurement boundary.

Index Terms—ECC, SECDED, Hsiao code, post-route power, switching activity, matched physical design

I. INTRODUCTION

ECC semantics determine correction and detection behavior, but not a unique circuit or physical energy. RTL organization fixes register placement and transaction timing; synthesis and place-and-route determine cells, wiring, and parasitics; an activity model determines which capacitances and cell states are exercised. Moving a conclusion across any one of these boundaries can change the quantity being claimed.

This matters for memory ECC because the logic sees structured events. Writes, clean reads, corrections, detections, and idle windows sensitize different cones. Transition-density and power-estimation studies established that signal statistics condition dynamic-power estimates [1], [2]. Ghosh *et al.* went further for memory ECC: they optimized Hamming and Hsiao check matrices using SPEC and MediaBench traces [3]. Their work rules out any broad claim that trace-aware ECC power analysis or Hamming/Hsiao activity sensitivity is new.

Our narrower question is whether a *winner* obtained from vectorless post-route power remains the winner after operation-specific activity is applied to the same physical states. That question requires more than functional labels. Earlier qualified experiments in this repository showed that exact-equivalent temporal SECDED implementations can move substantially in area, timing, latency, and energy, whereas an exact-equivalent flat/hierarchical Hsiao transformation produces a smaller mixed

displacement. Re-implementing the temporal pair at a second target changes effect magnitude. These observations motivate the controls used here: a test of activity identity must freeze hardware identity, implementation condition, routed state, feasibility status, and parasitic association.

We therefore compare conventional Hamming-style and minimum-odd-weight-column Hsiao SECDED logic [4] under one 10-ns SKY130HD flow and seeds {11, 13, 17, 19, 23}. Each final routed netlist is first timing-qualified. Vectorless power and final-gate-netlist VCD power then use that run's own final SPEF. The measured quantity is operation-normalized ECC-logic energy; SRAM-macro-internal energy is excluded.

The contributions are:

- 1) an evidence-qualified method that separates semantic, formal/functional, hardware, physical, timing, parasitic, and activity identities before admitting an energy comparison;
- 2) a paired post-route dataset of ten timing-feasible implementations, 46 qualified activity records, and 23 matched operation/seed comparisons, in which the Hsiao-lower vectorless ordering held in 5/5 seeds but survived in only 4/23 operation-specific comparisons; and
- 3) a controlled interpretation that connects the activity result to exact-equivalent temporal and structural transformations, condition sensitivity, component accounting, and a stronger-correction feasibility boundary without turning those controls into independent headline claims.

The primary DATE identity is D9 through activity-conditioned energy; D13 is secondary through post-route matching and evidence association.

II. RELATED WORK

Hsiao codes minimize odd-weight parity-check columns for SECDED [4]; later work targets delay or adjacent-error guarantees [5], [6]. Stronger-correction and memory studies explore latency, redundancy, density, and adaptive protection [7]–[11]. These works show why code construction and system context matter, but they do not make an implementation's energy independent of its activity source.

Ghosh *et al.* is the closest verified comparator [3]: it uses application memory traces and directly studies Hamming and Hsiao checker power. Our distinction is experimental rather than conceptual. We preserve each final routed architecture/seed state and its extracted parasitics, replace only the activity abstraction, normalize by completed operations, and test whether the induced ordering changes across matched seeds. In the literature set audited for this paper, we found no study of that complete

matched final-route ordering-substitution question; this is an awareness claim, not proof of universal absence.

OpenROAD and OpenLANE support inspectable RTL-to-layout studies [12], [13], while equivalence methods support qualification of transformations [14], [15]. Flow repeatability alone, however, does not prevent a timing report, VCD, and SPEF from referring to different physical states. Our methodology therefore treats their association as part of the claim.

III. EVIDENCE-QUALIFIED METHODOLOGY

A. Qualification boundaries

Figure 1 shows the evidence chain. Coding guarantees admit a candidate only at a declared boundary; the proof or test relation must then match the architecture pair. Hardware identity includes RTL, interface, latency, and initiation interval (II). Physical identity adds implementation condition, library/corner, flow, and seed. Timing feasibility, final parasitics, activity identity, and operation normalization are separate gates.

The exact-equivalence claims are deliberately pair-specific. For conventional SECDED, zero-plus-basis affine encoder qualification and compositional decoder qualification establish identical systematic (72,64) transaction behavior after shifting pipelined responses by two cycles. The external request latency is one versus three cycles and II remains one; encoder and decoder response/status fields are aligned. For flat versus hierarchical Hsiao, arbitrary-word SAT proves decoder equality and temporal induction proves same-cycle equality at the registered transaction boundary; both have one-cycle latency and II=1. Conventional SECDED versus Hsiao instead has distinct internal codeword representations. We qualify their common interface and W1-correction/W2-detection contract, and do not claim exact codeword equivalence. The complete wording matrix is retained with the artifact.

B. Physical and activity identity

All main-study runs use SKY130HD at the nominal typical corner, the same interface, 64-bit payload, physical-flow policy, constraints, and 10-ns target. Seeds 11, 13, 17, 19, and 23 are paired within architecture comparisons. They are deterministic perturbations of physical-design heuristics, not Monte Carlo process samples, fabricated-chip samples, or independent draws from a physical population. Pairing is important because an architecture delta can be comparable to placement and routing variation.

Each activity measurement uses the exact final zero-delay routed gate netlist associated with its timing result and that run's final SPEF. Hash joins bind architecture, seed, routed netlist, SPEF, VCD/workload, and power result. The vectorless diagnostic uses the flow's default activity assumptions. The operation-specific estimator uses five controlled classes: idle, clean write, clean read, correctable single-bit read, and detectable double-bit read. Sixteen warm-up cycles precede 256 measured operations. For architecture a , seed s , and operation o ,

$$E_{a,s,o} = P_{a,s,o} T_{s,o} / N_o, \quad N_o = 256. \quad (1)$$

The paired effect is $\Delta E_{s,o} = E_{\text{Hsiao},s,o} - E_{\text{conv},s,o}$; a negative value favors Hsiao. Power and energy magnitudes are not compared across estimators; only their matched architecture orderings are compared.

Every admitted record represents all 72 SRAM-output activity roots. Functional logic coverage is 99.298390–99.356061%, including 100% sequential and approximately 99.2% combinational coverage. Coverage qualifies annotation completeness, not workload representativeness. Seed 11 lacks qualified idle and write counterparts, so those classes have four pairs; no missing value is imputed. Means, sample standard deviations, ranges, and exact sign counts are descriptive. The matched physical pair is the statistical unit; no population significance claim is made.

The energy boundary is routed ECC logic around the memory interface. The open SRAM characterization does not provide the macro-internal switching energy needed for a whole-memory result. All numeric claims are therefore post-route ECC-logic estimates, not chip or silicon measurements.

C. Compact reproducibility contract

A record is admitted only after route completion, nonnegative setup WNS at 10 ns, final-netlist and final-SPEF association, VCD parsing, representation of all 72 output roots, coverage calculation, and successful component/energy extraction. The artifact package retains hashes, commands, generated data, scripts, full figures, and the two unavailable seed-11 pairs. Replication must rebuild the same identities or declare a new physical condition; a failed join makes a result unavailable rather than zero.

IV. EXPERIMENTAL DESIGN AND CONTROLS

The paper distinguishes three questions. *Q1–hardware identity*: can qualified RTL restructuring change physical outcome? *Q2–implementation condition*: can a target change modify effect magnitude? *Q3–activity identity*: can activity substitution change the architecture ordering when the routed state is fixed? Q3 is the principal experiment; Q1 and Q2 justify its controls.

The temporal control changes state placement, request latency, clock burden, and switching behavior together; it establishes why hardware identity cannot be transferred. The structural Hsiao control holds the matrix and temporal contract fixed and shows that a distinct implementation need not move by much. The 5-ns result belongs only to the temporal SECDED pair: it is not a 5-ns replication of conventional-versus-Hsiao operation energy. Finally, BCH has a different correction guarantee and is excluded from the 23 activity comparisons. The evaluated BCH(78,64, $t=2$) syndrome/Chien realization did not meet the common 10-ns target in the five matched implementation attempts. This is a categorical admission result for that realization, not a statement about BCH as a family.

V. RESULTS

A. Physical qualification and ordering substitution

All ten main-study implementations route and meet the 10-ns setup target. Across matched seeds, Hsiao minus conventional mean changes are $+237.6 \mu\text{m}^2$ total instance area,

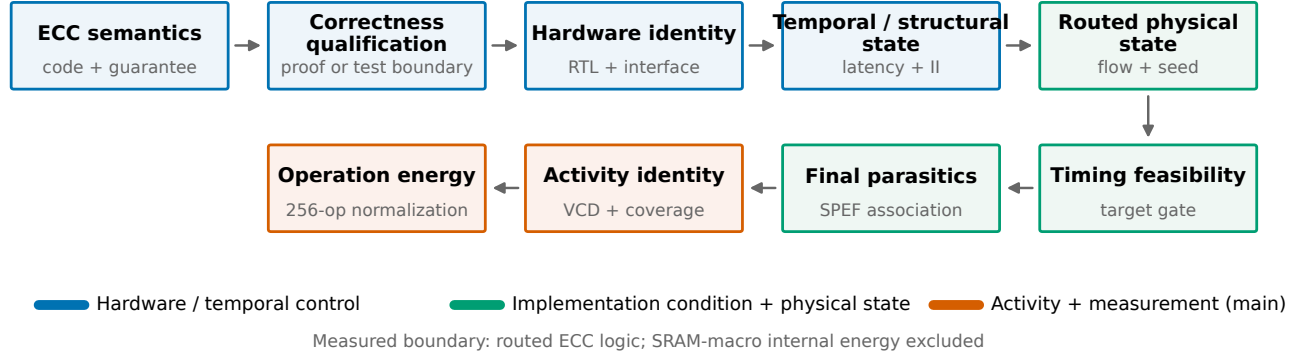


Fig. 1. Evidence chain used to isolate activity identity. Temporal/structural hardware, implementation condition, and activity are distinct dimensions. Controls establish the first two; the main experiment freezes them and substitutes the third.

TABLE I
CONTROLLED IMPLEMENTATION SPACE. CONTROLS ESTABLISH WHY HARDWARE AND CONDITION ARE FROZEN BEFORE THE MAIN ACTIVITY SUBSTITUTION. EFFECTS ARE CHANGED MINUS BASELINE, PAIRED WITHIN SEED.

Study	Change	Held fixed / qualification	Cond.; pairs	Main observed effect	Role here
Temporal SECEDED	combinational → pipelined	Same (72,64) transaction relation; exact alignment; II=1	10 ns; 5	Area +37.2%, timing +46.6%, energy/op −23.4%; directions 5/5; latency 1→3 cycles	Hardware identity control
Temporal SECEDED	same RTL pair under tighter target	Same qualification and pair identities	5 ns; 5	Area +36.7%, timing +68.6%, energy/op −19.3%; directions 5/5	Condition control; not activity replication
Structural Hsiao	flat → hierarchical decode	Same matrix, interface, latency=1, II=1; exact arbitrary-word and boundary proofs	10 ns; 5	Area −0.564%, wire +3.867%, vias +3.918%, timing +2.656%, energy +1.328%	Small mixed structural control
BCH feasibility	stronger correcting realization	W1/W2- Named syndrome/Chien RTL; common physical target	10 ns; 5	Evaluated realization timing-feasible in attempts	Categorical admission boundary
Main activity study	conventional SECEDED Hsiao ↔	Interface/reliability contract, 10-ns flow, matched physical identities	10 ns; 5	Vectorless favors Hsiao 5/5; operation-specific energy favors it 4/23	Principal experiment

+237.2 μm^2 standard-cell area, +2992.8 μm wire, +467.8 vias, and +0.2437 ns setup WNS. Hsiao has higher WNS in four of five pairs. Thus both architectures are admitted before energy is compared.

Figure 2 gives the headline result. Vectorless post-route power favors Hsiao in all five matched seeds. Operation-specific energy favors it in only 4/23 cells: 3/5 clean reads and 1/4 idle windows, with no Hsiao-lower write, correction, or detection pair. Because both estimators refer to the same qualified routes and parasitics, the mismatch is an activity-model ordering change rather than a different-implementation comparison.

B. Operation-specific energy

Figure 3 exposes all paired values. Mean Hsiao-minus-conventional energy is 0.0191 pJ for idle, +0.1207 pJ/write, −0.0308 pJ/clean read, +0.3095 pJ/correction, and +0.1935 pJ/detection. Clean read is the sole operation with a negative mean, but its seed range is −0.213 to +0.171 pJ/op. A single route could therefore support either read ordering. Correction has the largest mean displacement and favors conventional SECEDED in 5/5 pairs.

TABLE II
OPERATION-SPECIFIC MATCHED ENERGY ORDERING AT 10 NS.
 $\Delta E = E_{\text{Hsiao}} - E_{\text{conv.}}$

Operation	n	Mean ΔE (pJ/op)	Range (pJ/op)	Hsiao lower
Idle	4	+0.0191	−0.029→+0.068	1/4
Write	4	+0.1207	+0.073→+0.139	0/4
Clean read	5	−0.0308	−0.213→+0.171	3/5
Correction	5	+0.3095	+0.148→+0.521	0/5
Detection	5	+0.1935	+0.030→+0.405	0/5

For an illustrative active mix with nonnegative weights summing to one, the mean-delta model is

$$\Delta E = 0.1207w_{\text{write}} - 0.0308w_{\text{read}} + 0.3095w_{\text{corr}} + 0.1935w_{\text{det}}. \quad (2)$$

For read/write traffic alone it crosses zero at a 79.6476% clean-read fraction. This is an engineering interpolation of operation means, not a workload-trace result or a selector study.

Ordering changes when the activity abstraction changes ($\Delta = \text{Hsiao} - \text{conventional}$)

Seed 11	-281.94	—	—	+0.17	+0.52	+0.41
Seed 13	-331.40	+0.02	+0.07	-0.21	+0.16	+0.03
Seed 17	-330.82	-0.03	+0.14	-0.09	+0.24	+0.14
Seed 19	-330.08	+0.07	+0.14	+0.11	+0.48	+0.36
Seed 23	-299.19	+0.02	+0.13	-0.13	+0.15	+0.04
	Vectorless ΔP (μW)	Idle ΔE (pJ/op)	Write ΔE (pJ/op)	Clean read ΔE (pJ/op)	Correction ΔE (pJ/op)	Detection ΔE (pJ/op)

Blue: Hsiao lower Red: conventional lower —: unavailable (seed 11 idle/write)

Fig. 2. Matched Hsiao-minus-conventional ordering. Vectorless cells report μW ; operation cells report pJ/operation. Blue/negative favors Hsiao; red/positive favors conventional SECD. Numeric signs and labels make the figure grayscale readable. Missing seed-11 idle/write pairs remain visible.

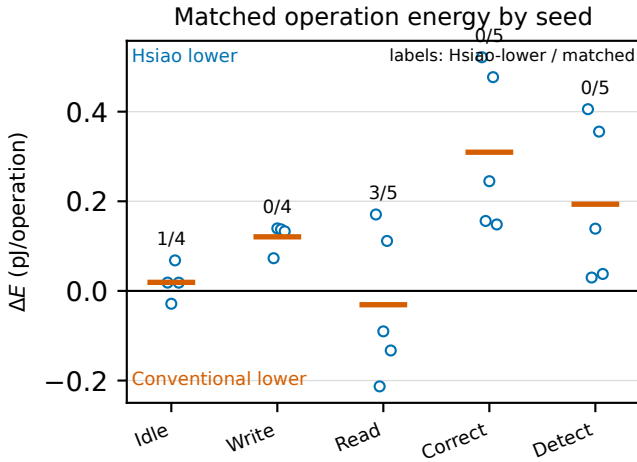


Fig. 3. All 23 matched operation-energy deltas. Open circles are seeds; short orange segments are arithmetic means. Counts give Hsiao-lower pairs over available pairs.

C. Component accounting

Figure 4 separates tool-reported internal, net-switching, and dynamic power, where dynamic is their sum; leakage is retained in the source table. For clean reads, Hsiao has lower net-switching power in 5/5 seeds (mean $-28.97 \mu\text{W}$) and higher internal power in 5/5 (mean $+25.89 \mu\text{W}$). The mean dynamic difference is $-3.08 \mu\text{W}$, leakage is $+0.00 \mu\text{W}$, and total energy is -0.0308 pJ/op .

Correction has mean internal, switching, dynamic, leakage, and total-energy differences of $+35.46 \mu\text{W}$, $-4.52 \mu\text{W}$, $+30.94 \mu\text{W}$, $+0.00 \mu\text{W}$, and $+0.3095 \text{ pJ/op}$. Detection gives $+30.42 \mu\text{W}$, $-11.07 \mu\text{W}$, $+19.34 \mu\text{W}$, $+0.00 \mu\text{W}$, and $+0.1935 \text{ pJ/op}$. Internal is higher in 5/5 correction and detection

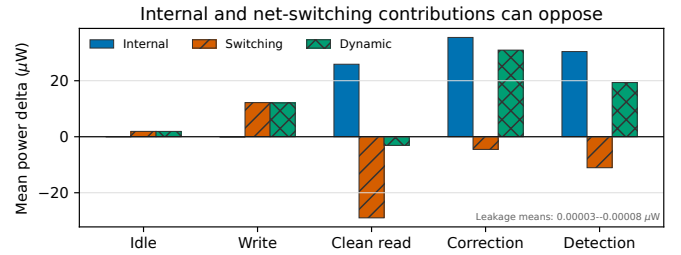


Fig. 4. Mean component deltas across available matched seeds. Dynamic is internal plus net switching; hatches and labels supplement color. Leakage is retained in source data and is negligible at this scale. The accounting is consistent with opposing contributions but does not establish a causal mechanism.

pairs; switching is lower in 3/5 and 4/5, respectively. These reports are consistent with internal and switching contributions competing after mapping and operation sensitization. They do not prove which net-level mechanism caused the totals.

D. What the controls establish

At 10 ns, pipelined minus combinational SECD changes standard-cell area by $+37.20\%$, the signed-slack-derived timing metric by $+46.63\%$, and equal-work steady-stream energy/op by -23.4190% ; directions hold in 5/5 pairs while latency changes from one to three cycles. Fresh 5-ns routes preserve the directions in 5/5 pairs but change the mean timing and energy effects to $+68.63\%$ and -19.3366% . This answers Q1 and Q2 only.

Hierarchical minus flat Hsiao produces a smaller mixed displacement: area -0.564% , detailed wire $+3.867\%$, vias $+3.918\%$, timing $+2.656\%$, and energy $+1.328\%$. Exact semantic and temporal identity does not eliminate physical displacement, but it does not prescribe a large effect. Implemen-

tation identity determines what must be measured separately; it does not predict the magnitude or sign of the resulting physical displacement.

VI. DISCUSSION, THREATS, AND CONCLUSION

A. One activity question, supported by controls

The three questions form one control argument rather than three papers. Q1 shows why a behaviorally qualified architecture change cannot borrow another implementation's PPA. Q2 shows why the implementation target belongs to physical identity. Those controls justify freezing both before Q3. Once frozen, replacing vectorless assumptions with operation-specific activity changes the observed Hsiao-lower ordering from 5/5 physical pairs to 4/23 operation/seed cells. The evidence rule is consequently stronger than an architecture verdict: report the implementation, physical condition, activity source, operation normalization, and measurement boundary together.

The controls are intentionally insufficient as stand-alone contributions here. Removing them would leave the 5/5-to-4/23 observation understandable, but would weaken the reason for fixing the exact route, temporal contract, target, and feasibility gate before substituting activity. Conversely, expanding their former plots or campaign histories would displace the experiment that answers Q3. Table I retains only the evidence needed to show what is controlled and why.

The temporal and structural results also prevent opposite overreadings. A large temporal displacement does not imply that every semantics-preserving RTL transformation matters equally; the sub-percent Hsiao area change and low-single-digit routing, timing, and energy changes provide a counterexample. A small structural displacement does not license pooling the two identities; several signs are consistent across all five pairs and one timing sign reverses. Implementation identity is therefore a measurement obligation, not an effect-size predictor.

The second timing target has the same limited role. Fresh 5-ns implementations show that target-dependent synthesis, buffering, placement, and routing can change the magnitude of a tradeoff even when the temporal pair's directions persist. Because no operation-specific conventional/Hsiao routes were generated at 5 ns, that evidence cannot answer whether Q3 repeats at a tighter target. It supports the decision to name the 10-ns condition, not a cross-condition activity claim.

B. Reading the ordering change

The vectorless result is not labeled inaccurate. It answers a different question under a default activity assumption, and its unanimous sign is a valid diagnostic observation for the five qualified routes. The operation-specific result conditions the same capacitances and parasitic state on explicit transactions. Their disagreement shows that the vectorless winner is not invariant to that substitution under the evaluated conditions; it does not show that either estimator is universally preferable.

The 4/23 count must likewise be read at the level at which it was formed. Each cell is a within-seed architecture pair for one operation, and the denominator excludes two unavailable

seed-11 pairs. The cells share hardware identities and operation definitions, so they are not 23 independent samples from a physical population. We use the count to describe how often an ordering survives in the complete matched table, not to estimate a Bernoulli probability, compute a confidence interval, or claim that Hsiao loses with some application-independent frequency.

Clean read illustrates why aggregate means are insufficient. Its mean favors Hsiao, yet two seeds favor conventional SECDED and the range crosses zero by more than the mean magnitude. Correction and detection have larger positive total-energy means even though their mean switching deltas are negative. The component table makes that accounting visible: net switching can decrease while internal power increases enough to change the dynamic and total signs. The evidence does not identify whether the internal term arose from a particular cell state, transition slew, path sensitization, or glitch sequence. Establishing such a cause would require delay-aware waveforms and per-cell or per-net attribution that were not retained here.

The read/write crossover has the same conditional status. It combines arithmetic means from controlled classes; it does not model temporal correlation, burst boundaries, idle residency, correction rate, macro activation, or an application trace. It can help an engineer see why a single average ranking is insufficient, but it should not be used to select an ECC without supplying measured workload weights and the missing memory-energy terms.

C. Fairness and admission

The conventional/Hsiao comparison is fair at the declared boundary, not identical in every internal property. Both satisfy the same external interface and SECDED contract, run through the same flow, meet the 10-ns target, and are paired by physical seed. Their different parity organization and internal representation are the intended architecture difference. Exact codeword identity is neither present nor claimed. This distinction is why formal qualification is explicit rather than collapsed into the phrase "functionally equivalent ECCs."

The BCH point illustrates a different boundary. A W2-correcting guarantee cannot be exchanged for SECDED merely to obtain a feasible energy value, and a route that misses setup cannot enter an equal-target energy ordering. Retaining the 0/5 outcome avoids silently dropping a stronger guarantee, but the named syndrome/Chien realization is only one point in a broad BCH architecture space. Alternative parallelism, pipelining, field arithmetic, constraints, or libraries could produce a different feasibility outcome.

Physical seeds have a similarly narrow interpretation. They expose whether a sign is peculiar to one deterministic optimization trajectory under a frozen flow. They do not sample process variation, placement distributions under randomized laboratory conditions, or manufactured devices. Five pairs can strengthen or weaken a within-flow sign observation; they cannot establish technology portability or statistical significance. Reporting all pairs rather than only a mean preserves that distinction.

D. Threats and falsification

Workload representativeness: controlled operation classes are not application traces. *Power model*: final-routed-gate-netlist zero-delay activity plus extracted parasitics omits some delay-induced glitches. *Memory boundary*: SRAM-macro-internal energy is absent. *Physical population*: five deterministic matched seeds do not define a probabilistic population. *Missing records*: seed 11 lacks qualified idle/write pairs. *Estimator dependence*: one vectorless baseline configuration is tested. *Generality*: the principal result uses one library, corner, data width, flow, and 10-ns activity-aware condition. *Formal boundary*: only the temporal SECDDED and structural Hsiao pairs have exact-equivalence evidence; the main code comparison uses interface/correction qualification. *Condition boundary*: the 5-ns result applies only to the temporal SECDDED control. *BCH boundary*: the feasibility result applies only to the evaluated syndrome/Chien realization.

The central result would be falsified if an evidence join associates a VCD or SPEF with the wrong route, if re-extraction changes a reported sign, if an admitted implementation fails the 10-ns gate, or if the regenerated 23-cell table does not contain the stated four negative deltas. The artifact therefore retains full-precision sources, hashes, generated tables, scripts, and unavailable cells. A new technology, route policy, activity source, or workload producing a different ordering would not invalidate this record; it would define a new physical or activity identity and test generality.

E. Conclusion

Under these conditions, a vectorless architecture ordering does not generally survive operation-specific activity. Component accounting shows that lower net switching can coexist with higher internal power and higher total operation energy. The supporting controls establish why temporal structure, implementation condition, and feasibility must be fixed before interpreting that substitution. Future work should repeat the ordering test with application traces, delay-annotated activity, macro energy, other corners and libraries, and prospectively defined physical populations. Until then, the defensible conclusion is conditional: ECC energy ordering is a property of a qualified implementation under a stated activity and measurement boundary, not of a code label alone.

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